MAYANK GANGREDIWAR

APPLIED ML/DL ENGINEER

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Pune, Maharashtra – 411044, India

OBJECTIVE

Applied AI engineer with experience across deep learning research, computer vision/OCR proof-of-concepts, multi-modal transformers, and production AI systems. Combines model evaluation and experimentation with the ability to design and build deployable AI backends. Focused on understanding model behaviour, improving task performance, and applying ML/DL to real-world problems.

SKILLS

Programming: Python, SQL, C++

ML & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, computer vision, YOLO, pretrained vision models, transformer architectures, model evaluation, ablation studies

GenAI & Retrival: RAG, embeddings, semantic search, vector databases, custom tool orchestration, Ollama, Amazon Bedrock, LlamaIndex

Backend / MLOps: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Pydantic, JWT/OAuth2/RBAC, MLflow, Langfuse, GCP/Cloud Functions, Git/GitHub

WORK EXPERIENCE

Associate AI Software Engineer Aug 2026 – Present
Sciverse Solutions *India*

- Lead AI-side architecture for a SaaS assistant, designing a custom layered orchestration framework with explicit component contracts to coordinate structured database tools and RAG retrieval; coordinate integration with back-end/RAG engineers and report progress to the solution architect.
- Implementing multi-round tool execution, JWT-based identity and role extraction, per-tool RBAC, token/cost-budget checks, structured response schemas, and post-response factuality/format validation.
- Built lightweight in-house guardrails and used locally hosted Ollama models for early development instead of paid cloud guardrail and inference services, avoiding external model/guardrail charges during this phase.
- Preliminary manual and scripted evaluation across approximately 50 scenario-based questions showed about **95% answer correctness**; RBAC-related behaviour remained under development during testing.

AI Product Development Intern Jun 2025 – Dec 2025
Asymmetrical Learning *India*

- Engineered and deployed **8+ Python AI microservices** integrating OpenAI, Gemini, Claude, Deepgram, and ElevenLabs through FastAPI and Google Cloud Functions.
- Co-developed Wriddles, an AI-powered personalised learning application deployed on Google Play and the Apple App Store; built LLM-driven content and metadata workflows and adaptive recommendations using explore–exploit algorithms.

PROJECTS

Wriddles: AI-Powered Personalized Learning Platform
Tech: Flutter, Firebase, Google Cloud Functions (GCP), Python, Gemini API

- Co-developed and deployed an AI-powered learning platform using Flutter, Firebase, GCP, and Gemini API, implementing explore–exploit recommendation algorithms for personalized learning, automated content classification, moderation, and metadata generation.

Employee Onboarding & Competency Platform
Tech: React, TypeScript, FastAPI, PostgreSQL, SQLAlchemy, Google OAuth, JWT, TanStack Query, TailwindCSS, Render, Vercel

- Built **20+ REST APIs**, a PostgreSQL schema with **15+ relational tables**, Google OAuth/JWT authentication, role-based workflows, and document-based quiz import.

EDUCATION

Bachelor of Technology, Electronics and Telecommunications, PCCOE, Pune (2022-2026, GPA 7.66)
Honors in Deep Learning, PCCOE, Pune (2024 - 2026, GPA 9.0)

PUBLICATIONS

- **A Modality-Aware Multimodal Transformer for Chromatin Accessibility Prediction.** Co-author; submitted to BMC Informatics (under review). Integrated DNA sequence, histone-modification, and transcription-factor signals; reported Pearson correlation of 0.802 and ROC-AUC of 0.890; conducted modality ablation studies.
- **AI-Based Smart Water Quality Monitoring and Disease Outbreak Forecasting System.** Gangrediwar M. et al., IEEE PuneCon 2025, published in IEEE Xplore.